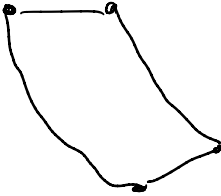


Teacher IS Pearson Pg 2

Questions

- 15.1) b ✓ 15.2) c ✓ 15.3) d ✓ 15.4) c ✓
15.5) b ✓ 15.6) d × c 15.7) c × b 15.8) d ✓
15.9) c ✓ 15.10) d ✓ 15.11) d ✓ 15.12) a × c
15.13) a ✓ 15.14) c ✓

a)



$$P_2 = 1 \cdot 10^5 \text{ Pa} \quad V_1 = 16 \cdot 10^{-3} \text{ m}^3$$
$$T_1 = 300 \text{ K} \quad V_2 = 0,8 \cdot 10^{-3} \text{ m}^3$$
$$T_3 = 1900 \text{ K}$$

a) 1 → 2:

$$P_1 V_1^\gamma = P_2 V_2^\gamma \rightarrow P_2 = P_1 \left(\frac{V_1}{V_2} \right)^\gamma = 6,63 \cdot 10^6 \text{ Pa}$$

$$T_1 V_1^{\gamma-1} = T_2 V_2^{\gamma-1} \rightarrow T_2 = T_1 \left(\frac{V_1}{V_2} \right)^{\gamma-1} = 994 \text{ K}$$